

1. The gradient is

$$\nabla f(x) = \begin{pmatrix} 100x_1 \\ -3x_2^2 \end{pmatrix}, \quad \nabla f(x^*) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Therefore, the first order necessary condition is verified.

The second derivative matrix is

$$\nabla^2 f(x) = \begin{pmatrix} 100 & 0 \\ 0 & -6x_2 \end{pmatrix}, \quad \nabla^2 f(x^*) = \begin{pmatrix} 100 & 0 \\ 0 & 0 \end{pmatrix}.$$

It is positive semidefinite. Indeed, the eigenvalues are read on the diagonal, as it is a diagonal matrix. And they are non negative. Therefore, the second order necessary condition is verified.

2. Note that the value of the objective function at x^* is

$$f(x^*) = 0.$$

Consider the direction

$$d_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

and the point x_1 obtained by following this direction from x^* with a step α :

$$x_1 = x^* + \alpha d_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + \alpha \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ \alpha \end{pmatrix}.$$

The value of the objective function at x_1 is

$$f(x_1) = -\alpha^3.$$

Therefore, for any $\alpha > 0$, we have

$$f(x_1) < f(x^*),$$

and x^* cannot be a local minimum.

Consider now the direction

$$d_2 = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

and the point x_2 obtained by following this direction from x^* with a step α :

$$x_2 = x^* + \alpha d_2 = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + \alpha \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ -\alpha \end{pmatrix}.$$

The value of the objective function at x_2 is

$$f(x_2) = \alpha^3.$$

Therefore, for any $\alpha > 0$, we have

$$f(x_2) > f(x^*),$$

and x^* cannot be a local maximum.